3.4.2 Mensuration and calculation

G14

Basic foundation content	Additional foundation content	Higher content only
use standard units of measure and related concepts (length, area, volume/capacity, mass, time, money etc.)		

G15

Basic foundation content	Additional foundation content	Higher content only
measure line segments and angles in geometric figures, including interpreting maps and scale drawings and use of bearings		

Notes: including the eight compass point bearings and three-figure bearings.

G16

Basic foundation content	Additional foundation content	Higher content only
know and apply formulae to calculate: area of triangles, parallelograms, trapezia; volume of cuboids and other right prisms (including cylinders)		

G17

Basic foundation content	Additional foundation content	Higher content only
know the formulae: circumference of a circle = $2\pi r = \pi d$ area of a circle = πr^2 calculate perimeters of 2D shapes, including circles areas of circles and composite shapes	surface area and volume of spheres, pyramids, cones and composite solids	

Notes: including frustums.

Solutions in terms of π may be asked for. See also N8, G18

G18

Basic foundation content	Additional foundation content	Higher content only
	calculate arc lengths, angles and areas of sectors of circles	

Notes: see also N8, G17

G19

Basic foundation content	Additional foundation content	Higher content only
	apply the concepts of congruence and similarity, including the relationships between lengths in similar figures	including the relationships between lengths, areas and volumes in similar figures

Notes: see also R12

G20

Basic foundation content	Additional foundation content	Higher content only
	know the formulae for: Pythagoras' theorem, $a^2 + b^2 = c^2$ and the trigonometric ratios, $\sin\theta = \frac{opposite}{hypotenuse}$, $\cos\theta = \frac{adjacent}{hypotenuse}$ and $\tan\theta = \frac{opposite}{adjacent}$	
	apply them to find angles and lengths in right-angled triangles in two dimensional figures	apply them to find angles and lengths in right-angled triangles and, where possible, general triangles in two and three dimensional figures

Notes: see also R12

G21

Basic foundation content	Additional foundation content	Higher content only
	know the exact values of $\sin\theta$ and $\cos\theta$ for $\theta = 0°, 30°, 45°, 60°$ and $90°$ know the exact value of $\tan\theta$ for $\theta = 0°, 30°, 45°, 60°$	

Notes: see also A12

G22

Basic foundation content	Additional foundation content	Higher content only
		know and apply the sine rule, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ and cosine rule, $a^2 = b^2 + c^2 - 2bc\cos A$ to find unknown lengths and angles

G23

Basic foundation content	Additional foundation content	Higher content only
		know and apply Area = $\frac{1}{2}ab\sin C$ to calculate the area, sides or angles of any triangle